CS5298: BLOCKCHAIN AND WEB3.0 SECURITY

New Syllabus Proposal

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Blockchain and Web3.0 Security

Subject Code

CS - Computer Science

Course Number

5298

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

(CS2310 Computer Programming or equivalent)
AND (CS3103 Operating Systems or equivalent)
AND (CS3201 Computer Networks or equivalent)

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course focuses on blockchain technologies and Web3.0 security. We start with the fundamentals of cryptography and distributed systems that underpin modern blockchain platforms, including collision-resistant hash functions, digital signatures and classical consensus algorithms in trusted setups (e.g., Paxos). In the next part, we focus on decentralized and untrustworthy environments, discussing permissioned blockchain systems (e.g., Hyperledger Fabric and Tendermint) and permissionless blockchains (e.g., Bitcoin and Ethereum). In the last part of our class, we introduce novel applications enabled by blockchain technology, including smart contracts, exchanges, and DeFi.

Course Intended Learning Outcomes (CILOs)

	CILOs	Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Understanding and mastering blockchain related concepts and techniques in different domains such as cryptography and consensus.	40	x	x	
2	Analysing the design of existing permissioned and permissionless blockchain systems.	35	x	x	
3	Developing and implementing blockchain system prototypes as well as blockchain-based applications.	25	x	x	

A1: Attitude
Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability
Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments
Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

	LTAs	Brief Description	CILO No.	Hours/week (if applicable)
1	Lectures	The fundamentals of cryptography and distributed systems that underpin modern blockchain platforms will be discussed. Principles, techniques and technologies used in today's permissionless blockchains as well as permissioned blockchains will be discussed. Novel applications enabled by blockchain infrastructure will be also introduced, including smart contracts and decentralized finance.	1, 2, 3	2 hours / week
2	Tutorials	Tutorials will be conducted in laboratory in the forms of discussion, demonstration, and hands-on sessions. Students will work with selected blockchains and ecosystem tools. This provides students with hands-on experience in analyzing the vulnerabilities and complexity of blockchain consensus algorithms. Students will also develop blockchain prototypes as well as tools to analyze and interact with them in Python. Students will also get hands-on experience in writing smart contracts in Solidity.	1, 2, 3	1 hour /week

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("-" for nil entry)	Allow Use of GenAI?
1	Individual Assignment 1	1, 2	15	-	No
2	Project 1	2, 3	15	-	Yes
3	Project 2	1, 3	20	-	Yes

Continuous Assessment (%)

50

Examination (%)
50

Examination Duration (Hours)
2

Minimum Examination Passing Requirement (%)
30

Additional Information for ATs
For a student to pass the course, at least 30% of the maximum marks of the examination must be obtained.

Assessment Rubrics (AR)

Assessment Task
Individual Assignment 1 (including problem set and hands-on exercises)

Criterion
Ability to understand design principles behind major consensus protocols and blockchain systems.

Excellent
(A+, A, A-): High

Good
(B+, B, B-): Significant

Fair
(C+, C, C-): Moderate

Marginal
(D): Basic

Failure
(F): Not even reaching marginal levels

Assessment Task
Project 1

Criterion
Ability to implement a Byzantine fault-tolerant protocol (a variant of PBFT) inside the open source Bedrock platform.

Excellent
(A+, A, A-): High

Good
(B+, B, B-): Significant

Fair
(C+, C, C-): Moderate

Marginal
(D): Basic

Failure

(F): Not even reaching marginal levels

Assessment Task
Project 2

Criterion
Ability to build a basic cryptocurrency exchange in Python to trade Bitcoin for Ether.

Excellent
(A+, A, A-): High

Good
(B+, B, B-): Significant

Fair
(C+, C, C-): Moderate

Marginal
(D): Basic

Failure
(F): Not even reaching marginal levels

Assessment Task
Examination

Criterion
The exam will include questions to assess the student's ability to explain the principles, techniques and technologies used in blockchains.

Excellent
(A+, A, A-): High

Good
(B+, B, B-): Significant

Fair
(C+, C, C-): Moderate

Marginal
(D): Basic

Failure
(F): Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

The syllabus evolves over time as security topics change. The following is a list of potential course topics:

(1) Traditional consensus: State Machine Replication, Paxos

(2) Consensus in Untrustworthy Environments: PBFT, HotStuff, Zyzzyva
(3) Permissionless Blockhains: Proof of Work, Proof of Work Challenges, Proof of Stake, Layer-2 Solutions, Cross-Chain Transactions
(4) Permissioned Blockchains: Different paradigms and systems including HyperLedger Fabric, Quorum, Tendermint
(5) Smart Contracts: Ethereum Smart Contracts, Solidity, Fungible & Non-Fungible Tokens
(6) Exchanges: Cryptocurrency Exchanges, DEXs, Market Algorithms
(7) DeFi: Decentralized Lending, Stablecoins, DeFi Security

Reading List

Compulsory Readings

	Title
1	-

Additional Readings

	Title
1	An Introduction to Mathematical Cryptography (Jeffrey Hoffstein, Jill Pipher, and J. H. Silverman)
2	Blockchain-Enabled Large-Scale Transaction Management (Mohammad Javad Amiri, Divyakant Agrawal, Amr El Abbadi)